AirCopBench: A Benchmark for Multi-drone Collaborative Embodied Perception and Reasoning

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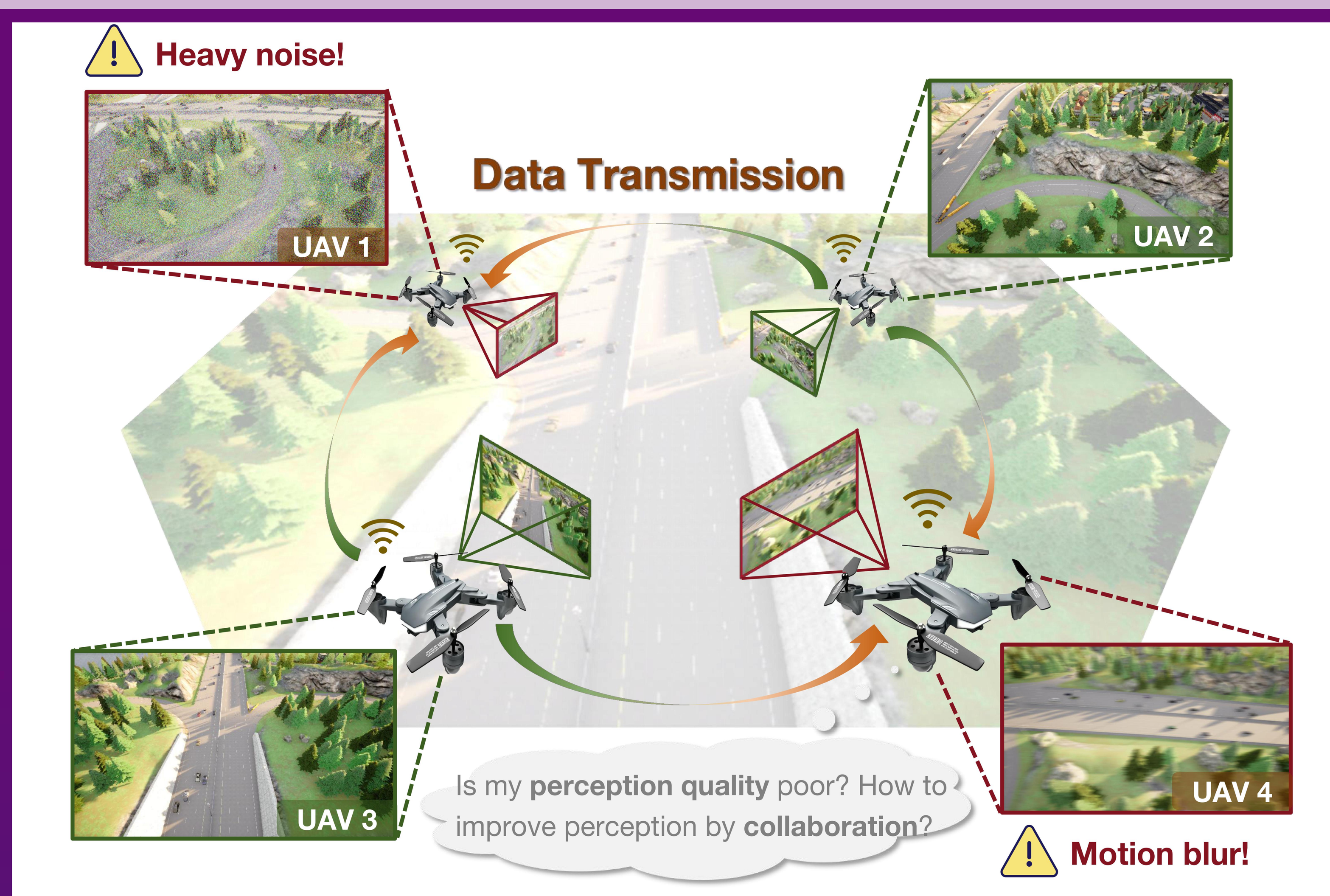
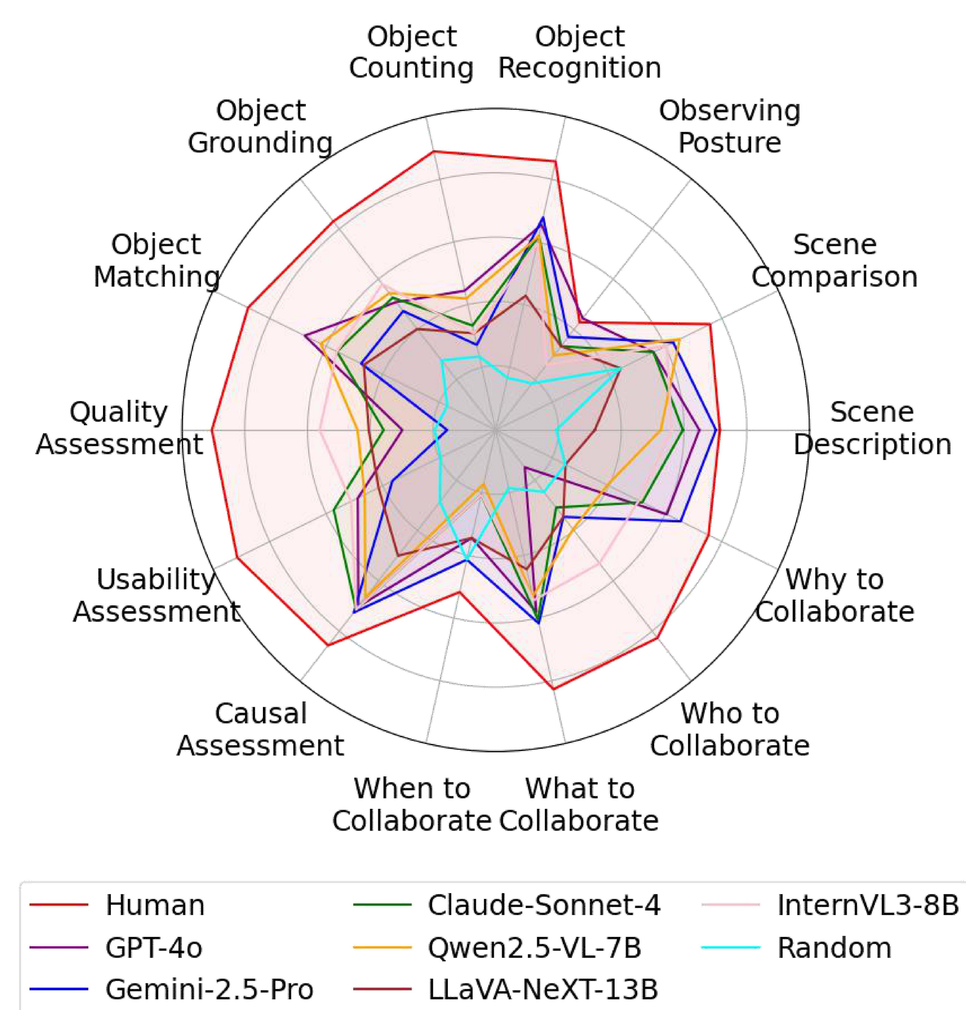
1. Introduction: MLLM for Collaborative Perception and Reasoning

➢**Motivation.** Although *multi-drone systems* provide superior coverage, robustness, and coordination, existing benchmarks largely *evaluate MLLMs* only in single-agent vision settings. Most *multi-image benchmarks* focus on perception with high-quality images, failing to capture the complexity of *egocentric collaboration and real-world degraded conditions*.

➢**Contribution.** The *first benchmark* for semantic embodied aerial collaborative perception and reasoning.





2. Task Setting: 4 Dimensions and 14 Types

Scene Understanding

1. Scene Description
Q: What is the dominant environmental feature in UAV1's view?
A: A urban road with a green grassy area.

2. Scene Comparison
Q: Which UAV perspective has more vehicles? A: UAV2.
UAV1: 5 cars; UAV2: 13 cars

3. Observing Posture
Q: Which perspective is closer to the ground?
A: UAV1 is closer with a clearer view.

Object Understanding

4. Object Recognition
Q: How many types of objects are there? A: 2 types.

5. Object Counting
Q: How many vehicles are visible on the road? A: 5 cars.

6. Object Grounding
Q: Where is the white van?
A: The white van is in a different lane behind the red car.

7. Object Matching
Q: Which object in UAV2 is the person standing near the traffic light in UAV1? A: The person in white standing by the red-and-white sidewalk edge in UAV2.
matched pedestrian

perception area · data link · How to enhance my perception?

Perception Assessment

8. Quality Assessment
Q: How would you rate the overall image quality?
A: Good with minor blur.

9. Usability Assessment
Q: Is this image usable for object detection tasks?
A: Moderately usable with some occlusions.

10. Causal Assessment
Q: What is the main factor that affects the object detection performance of this image?
A: Partial occlusion by trees.
occluded red car

Collaborative Decision

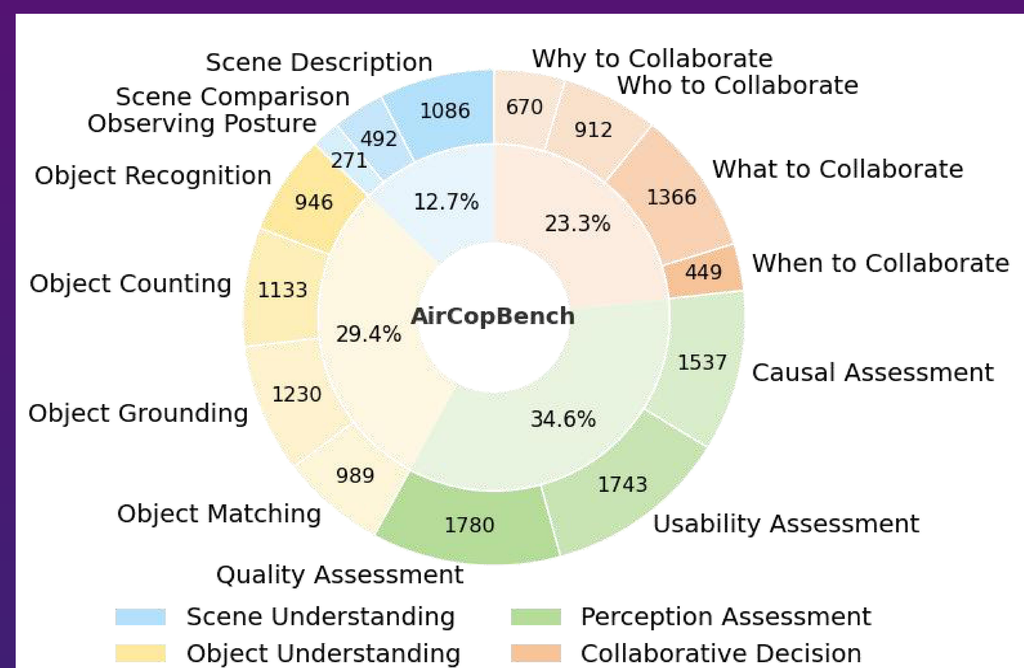
11. When to Collaborate
Q: Should UAV2 communicate with others for more info now?
A: Yes.
complete red car

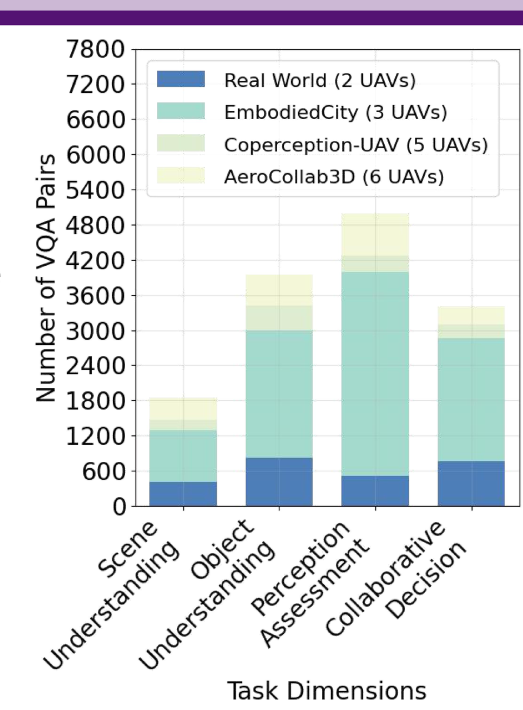
12. What to Collaborate
Q: What info should UAV1 share with UAV2?
A: Details of the red car's position and movement.

13. Who to Collaborate
Q: Which UAV should UAV2 collaborate with?
A: UAV1.

14. Why to Collaborate
Q: Why should UAV2 collaborate with UAV1?
A: To overcome partial occlusion of the red car.

3. Dataset Statistics





- **2.9k+** multi-view images (simulator + real-world + derived data)
- **14.6k+** visual question answering (VQA) pairs (14 sub-tasks)
- **9** perception degradation scenarios (occlusion, shadow, lighting, blur, noise, loss, etc.) and **4** UAV collaboration configurations

5. Error Analysis

Spatial Reasoning Error — Confused "opposite" and "ahead".

Question: What is the relative position of the white car compared to the bus?
A. The white car is driving in the adjacent lane directly **opposite** the bus. GT
C. The white car **is ahead of** the bus in the adjacent lane. MLLM

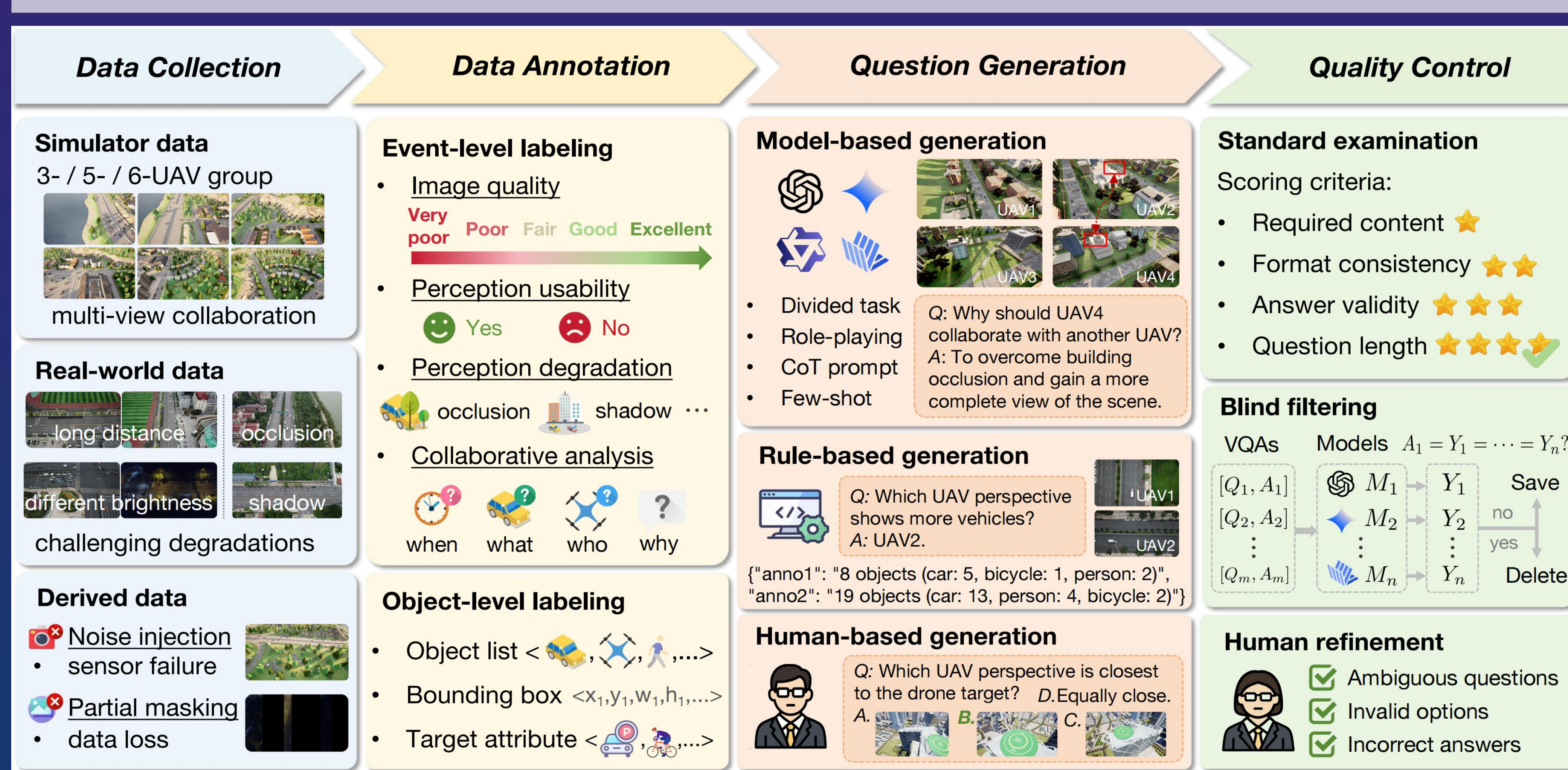
Perception Hallucination Error

Question: Which type of vehicle is most frequently observed in the parking lot?
A. **SUVs** MLLM
B. **Compact cars** GT
Misclassifying object due to perceptual bias in visual features.

Multi-image Understanding Error — Mismatching motion states.

Question: Which object in UAV2 corresponds to the dark gray car approaching the intersection from the left in UAV1? MLLM
A: The dark gray car now seen **turning right** at the intersection in UAV2. GT
C. The dark gray car now **approaching** the intersection from the top in UAV2.

4. Dataset Generation



6. Evaluation Results

Performance Analysis We evaluate **40** popular MLLMs. The *best-performing* models achieve only **59.17%** accuracy, **24.38%** below human performance, emphasizing the challenges in multi-view collaborative perception and reasoning. A pronounced *task bias* emerges, with strong performance on scene description and comparison but poor results on usability assessment and collaborative timing, while several *open-source models* rival or surpass GPT-4o.

SFT and Sim-to-Real After *fine-tuning*, the models' performance improved by up to **26.97%**. Moreover, fine-tuned MLLMs trained on simulated data show notable *sim-to-real* improvements in real-world tasks.